

Haoran Feng

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EDUCATION

Sep 2025 - Jun 2027 (Expected)	CUHK-Shenzhen	M.S., Integrated Circuits and Systems
Sep 2021 - Jun 2025	Shandong University (985)	B.Eng., Electronic Science / Computer Science

Honors: Academic Scholarship (Top 20%); Special Talent Scholarship in Innovation, Shandong University

INTERNSHIP EXPERIENCE

Apr 2026 - Present	Ubiquant Investment - AI Lab	LLM and HPC Intern
FlashInfer CUDA Graph reliability: Reproduced a vLLM TP=2 rollout hang on H200 and traced it to FTZ making a floating-point check misclassify valid negative subnormal FP32 values as the Lamport -0.0 sentinel in fused AllReduce + RMSNorm polling. Switched to an exact 0x80000000 bit check; two-GPU and model graph on/off regressions passed with no further hangs. R3 Router Replay: Built route capture, transport, and replay between vLLM and Megatron with response-mask alignment and 18 CPU tests. On 8xH200/Qwen3-30B-A3B, reduced route mismatch from 17-19% to 0, F-tau2 by 36-145x, and KL by 4-7x; scaled the workflow to 128 GPUs for a 200-step internal 300B run. Deterministic Router GEMM on H200: Added Triton full-K and DeepGEMM SM90 BF16 paths for decode-time small-M workloads. Across 20 model scenarios, DeepGEMM reduced median latency for 48 router GEMMs from 789.23 to 210.01 us/step (-73.39%) and total GPU busy time by 6.16%, with 135/135 kernel and 20/20 bitwise model checks passing. Asynchronous OE operator: Kept token-history processing on GPU and fused hashing in Triton, eliminating host synchronization across prefill, decode, mixed batches, and state recovery. Passed 28/28 TP1 cases with +4.99% throughput; restored TP2/TP4 determinism to 84/84 cases with 0 mismatches and improved decode throughput by about 4.6%/3.0%.		
Jan 2026 - Apr 2026	Moore Threads	Operator & Compiler Optimization Intern
TensorFlow MUSA Extension: Added operator support, plugin integration, tests, and graph optimizations. Integrated muDNN GELU and fixed the fusion pipeline so all 11 GELUs in the target network were fused, cutting representative-shape latency by 36.6%. OOM and crash fix: Traced long-run failures to shape tensors entering the device path in StridedSlice<int32>/Pack<int32>. Reworked HostMemory handling, improving success from about 30% at 500 iterations to 100% at 1,000 and passing 40K/400K/800K-iteration stress runs.		

OPEN-SOURCE CONTRIBUTIONS

Aug 2026	AI Infrastructure Upstream Contributions
Mooncake PR #3604 / #3601 (Merged): Fixed transfer stalls and timeouts caused by reusing stale QPs after RDMA port recovery. Reordered storage writes to call fdatsync before metadata commit and added regression coverage for sync failures and orphan-file cleanup.	

PROJECTS AND RESEARCH

Dec 2025 - Present

High-Performance Quantized LLM Runtime with PyTorch Extensions

- **Runtime and backend integration:** Extended a PyTorch runtime supporting W4A16 AWQ, W8A8 SmoothQuant, and FP8 with configurable FlashAttention-2/4, GQA KV-head expansion, soft-cap masking, and an SDPA fallback. Made CUDA architecture, cache, and model paths configurable for local RTX 5060 and H200 deployment.
- **H200 kernels and tensor parallelism:** Ported AWQ to 2xH200 and used Compute Sanitizer to trace an out-of-bounds decode access in `gemv_kernel_g128` to reading eight scale/zero groups when `K/group_size=7`. Added bounds checks, current-stream launches, and error handling; all memcheck/initcheck/synccheck/racecheck cases passed. Implemented packed-AWQ TP=2 with Q/K/V and gate/up column sharding, O/down row sharding, and NCCL all-reduce.
- **32B deployment:** Quantized and deployed Qwen2.5-Coder-32B W4A16 on two GPUs. Reduced checkpoint size from 61.04 to 18.02 GiB (-70.5%), increased end-to-end output throughput from 2.53 to 4.47 tok/s (+76.8%), and cut peak memory from 32.02 to 10.63 GiB/rank (-66.8%). Validated with ten CUDA/PyTorch numerical cases, cross-rank token parity, and interactive prompts.

Jan 2026 - Present

[Token-Weighted CoT Distillation | AAI 2027 Submission](#)

- **Method and role:** Trained and tuned models for a three-stage pipeline: structure recovery, token-weighted supervision, and preference optimization. Estimated token importance by perturbing each teacher rationale token and measuring the drop in reference-answer likelihood, then applied weighted SFT and an auxiliary loss at high-importance positions.
- **Results and submission:** Ran LoRA/DPO training and vLLM TP=4 evaluation. Qwen2.5-7B-Instruct reached 94.01%/94.00% accuracy on GSM8K/SVAMP, exceeding the strongest baselines by 5.51/10.10 percentage points. Revised manuscript submitted to AAI 2027.

SKILLS

Focus and languages: RL infrastructure and LLM training/inference optimization; IELTS 6.5, CET-6, Huawei HCIA-AI certified.

Programming and systems: C/C++, Python, CUDA, Triton, PyTorch Extensions; vLLM, veRL, Megatron-LM, FlashInfer, CUTLASS, and NCCL.

Engineering: Cross-repository debugging, minimal reproducers, controlled experiments, automated testing, and benchmarking; profiling and correctness validation with Nsight Systems, Nsight Compute, and Compute Sanitizer.